

The Toronto Raccoon Activity Index*

A Serious Exploration of a Joke Dataset

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October 1, 2026

I'll write it

1 Introduction

The raccoon (*Procyon lotor*) is a mammal native to North America. While predominately located on the East coast and the Great Lakes regions, raccoon populations have spread rapidly across the continent since the introduction of European colonists and the fur trade. Notably, they have found considerable success in metropolitan areas, due to factors such as readily available food sources like waste bins and gardens, accessible habitats in housing and buildings, and a lack of natural predators. However, raccoons also pose risks to their neighbours, as they commonly carry diseases such as rabies and *Baylisascaris procyonis*, both of which cause severe neurological harm in humans. Moreover, raccoons nesting often cause dangerous and costly property damage.

In Toronto, the raccoon is a divisive figure. The “raccoon problem” is one that has plagued Toronto City Council, with a number of mayors investing in methods of population control, “raccoon-proof” garbage bins, and even floating the possibility of extermination programs. Conversely, Conrad - a dead raccoon left on a sidewalk for fourteen hours that became a national media sensation - was memorialised with a heritage plaque on the corner of Yonge and Church on the eve of his tenth anniversary in 2025. The raccoon is seen as a symbol of adaptability, resilience, and part of life in the Greater Toronto Area (GTA) - so much so that City of Toronto’s Open Data team released a Toronto Raccoon Activity Index (TRAI) dataset on April 1, 2026, describing the raccoon as a “prominent feature of Toronto’s urban environment” that necessitated detailed, hourly observations across the 25 wards of the city. While the TRAI is ostensibly a joke and likely randomly generated, its mere existence symbolises the complex relationship between the raccoons and the citizens of Toronto.

*View the code and datasets at <https://github.com/shadowmisstep/yyz-raccoon-activity>

This paper seeks to analyse the TRAI through two lenses - as a legitimate dataset of raccoon behaviour across the city, and as a deliberately created product that may reveal underlying assumptions and biased understandings of Toronto’s neighborhoods.

2 Data

The two datasets utilised in this paper were obtained through the OpenDataToronto Library - the Toronto Raccoon Activity Index, and the 2021 One Variable sheet from the Ward Profiles (25-Ward Model) dataset. Data was extracted using R and the `opendatatatoronto` library, and cleaned using the `tidyverse` package.

2.1 The Toronto Raccoon Activity Index

2.1.1 Description

The TRAI details hourly raccoon appearances across the 25 wards of the GTA from January 1 to March 30 of 2024 (Table 1). Additionally, the TRAI records the number of compromised green compost bins, average raccoon weight in kilograms, and the raccoon’s “confidence.” Given that there is no given methodology or definition for characterising raccoon behaviour, this metric was not used in analysis.

Table 1: Sample Raccoon Activity Across 5 Wards of Toronto January 1, 2024 00:00

Ward	Units Observed	Bins Compromised	Raccoon Confidence	Average Weight (kg)
Davenport	20	13	10.0	6.06
Spadina-Fort York	16	10	10.0	4.85
University- Rosedale	19	14	10.0	6.21
Toronto-St. Paul’s	12	7	7.9	6.05
Toronto Centre	18	11	10.0	6.48
Toronto-Danforth	19	14	10.0	5.19

Alongside raccoon behaviour, the TRAI records human-raccoon interactions (Table 2). These interactions are defined as confrontations (described as “standoffs”) with the number of incidents, average duration of the interaction in seconds, proximity to a patio, and complaints filed are recorded.

Table 2: Raccoon-Human Interactions in Toronto-Danforth January 1, 2024 from 00:00 to 06:00

Hour	Human Raccoon Confrontations	Average Standoff Duration (s)	Patio Proximity	Complaints Logged
0	4	31	FALSE	2
1	1	25	FALSE	2
2	3	16	FALSE	1
3	2	26	FALSE	2
4	0	0	FALSE	0
5	0	0	FALSE	0

2.1.2 Cleaning and Aggregation

This data can be analysed across a number of categories, such as creating an average of raccoon observations per hour, or the daily average raccoon appearances in the quarter. The **aggregate** function of the **stats** package is used to create these averaged datasets (Table 3).

Table 3: Daily and Hourly Averages of Observed Raccoons Across Toronto from January to March 2024

(a) Daily Average		(b) Hourly Average	
Date	Raccoons Observed	Hour	Raccoons Observed
2024-01-01	5.72	0	15.05
2024-01-02	5.68	1	15.11
2024-01-03	5.77	2	15.04
2024-01-04	5.71	3	15.06
2024-01-05	5.72	4	2.60
2024-01-06	7.56	5	2.63

2.2 Ward Profiles

2.2.1 Description

The Ward Profiles dataset is based on the 2021 Census of Population conducted by the Government of Canada. The dataset records the demographics for the population and households of each ward, including ethnocultural origins, language, education status, and employment. See Table 4 for a sample of how the data is structured.

Table 4: Types of Household Dwellings of Toronto and Sample Wards

Dwelling Structure	Toronto	Ward 1	Ward 2	Ward 3	Ward 4	Ward 5
Single-detached house	270350	11270	21860	18585	8855	13190
Semi-detached house	73160	1425	875	1365	4215	3315
Row house	63170	3645	2005	2340	1175	1265
Apartment or flat in a duplex	46470	3790	785	1500	2120	1845
Apartment in a building that has fewer than five storeys	162260	2015	1925	11100	14040	7105
Apartment in a building that has five or more storeys	542625	15980	17860	30645	18955	18300
Other single-attached house	2760	10	25	35	70	35
Movable dwelling	95	0	0	0	15	0

2.3 Cleaning and Merging

As the TRAI and the Ward Profiles share the same ID, each ward’s demographics can be compared with corresponding raccoon behaviour. The Ward Profiles dataset includes average and median values for income and average household size, which are then isolated and merged with existing raccoon behaviour datasets.

3 Results

3.1 City-Wide Behaviour

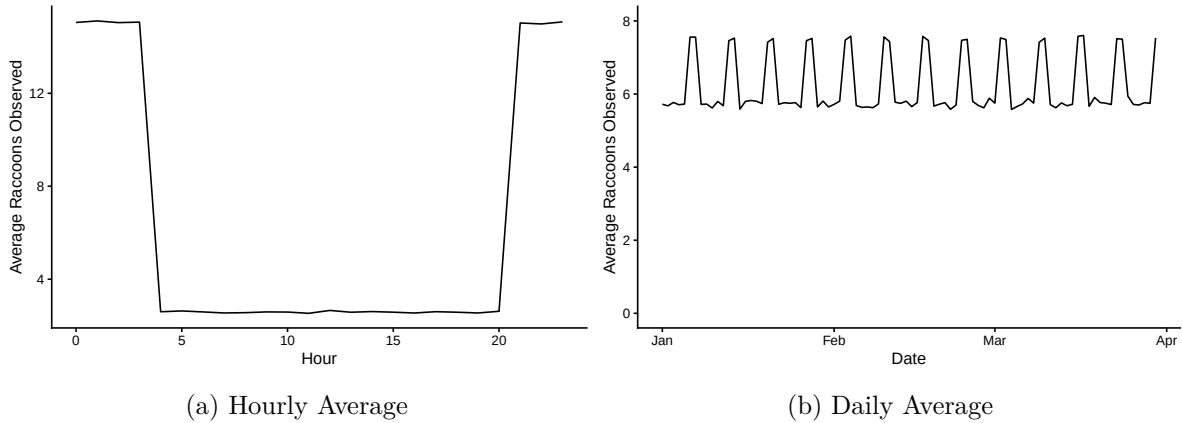


Figure 1: Average Raccoon Appearances Across Toronto

The TRAI featured a total of 54 000 observations over the timespan of the dataset. Overall, a total of 335 957 observations of raccoons were recorded. Raccoons observations peaked from the hours of 20:00 to 4:00, corresponding with their nocturnal nature, and around three to four times a month, likely corresponding to the waste management schedule for green bins, which are collected once a week (Figure 1).

Moreover, a higher presence of raccoons and number of observations directly correlates with higher raccoon activity, including compromised bins (Figure 2) and interactions with humans. These interactions also directly correlate to logged complaints, which reflects the combative nature of the relationship between residents and raccoons. These relationships are detailed in Figure 3.

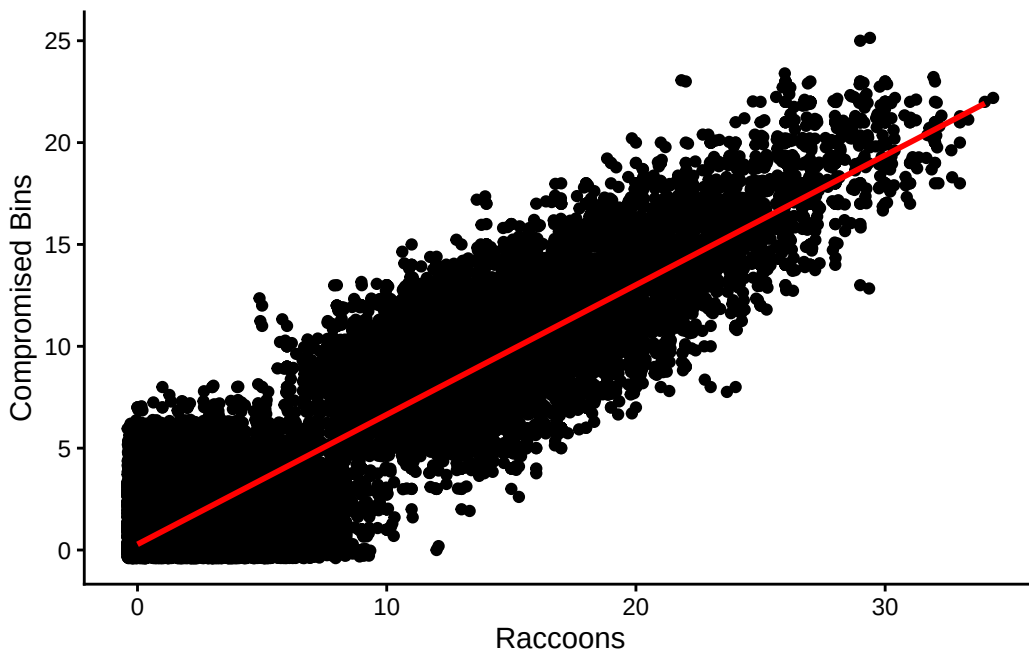
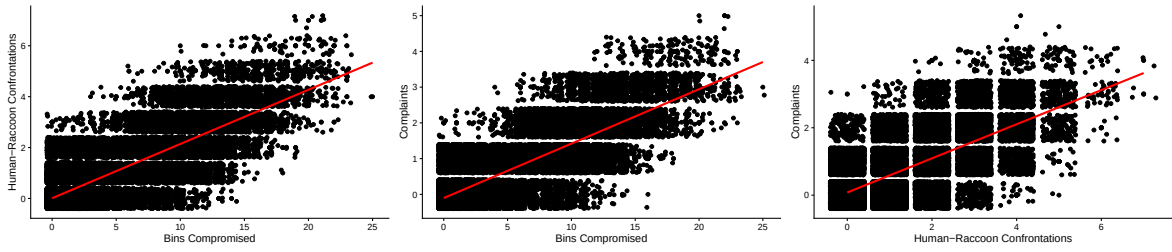


Figure 2: Raccoons Observed Compared to Number of Green Bins Compromised

3.2 By Ward

- Weight boxplot (who are outliers?)
- Wardly average raccoon observations bar graph (who are outliers?)
- Are the outliers between weight and observations the same?
- Total complaints per ward (outliers?)
- Total confrontations per ward (outliers?)
- plot demos (group bar?)
- Are there correlations?



(a) Number of Compromised Green Bins to Human-Raccoon Incidents
 (b) Number of Compromised Green Bins to Submitted Complaints
 (c) Human-Raccoon Incidents to Submitted Complaints

Figure 3: Human Responses to Raccoon Activity (with Jitter)